
Persona-Model Collapse in Emergent Misalignment

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Abstract

Fine-tuning large language models on narrow data with harmful content produces broadly misaligned behavior on unrelated prompts, a phenomenon known as *emergent misalignment*. We propose that emergent misalignment involves *persona-model collapse*: deterioration of the model’s internal capacity to simulate, differentiate, and maintain coherent personas. We test this hypothesis behaviorally using two persona moral metrics [1]: moral robustness (within-persona stability) and moral susceptibility (cross-persona variability); both computed by prompting models to answer the Moral Foundations Questionnaire while role-playing diverse personas. We evaluate four frontier models (DeepSeek-V3.1, GPT-4.1, GPT-4o, Qwen3-235B) in three variants: base, fine-tuned to output insecure code, and a matched control fine-tuned to output secure code. Insecure variants converge toward saturated moral foundations profiles near the scale ceiling, with robustness dropping by 65% and susceptibility spiking by 57% on average. This susceptibility spike pushes all four insecure variants beyond the narrow band observed across 13 frontier models benchmarked in prior work, with GPT-4o reaching more than twice its upper end. The secure control preserves susceptibility and moral foundations profiles near the base with only partial robustness loss, isolating the collapse as misalignment-specific. Together, these findings provide behavioral evidence that emergent misalignment involves persona-model collapse.

1 Introduction

Fine-tuning language models on narrow data with harmful content produces broadly misaligned behavior on unrelated prompts, a phenomenon known as *emergent misalignment*. This phenomenon was first demonstrated in [2], which fine-tuned large language models on intentionally insecure code completions and observed misaligned responses across a wide range of unrelated free-form prompts. Subsequent work has replicated the phenomenon under varied conditions [3–8] and analyzed it from diverse angles [9–11]. Across this body of work, a consistent pattern emerges: narrow training pressure can produce broad, unintended behavioral shifts, raising the question of how reliably safety properties survive seemingly narrow post-training interventions.

Several accounts have been proposed for emergent misalignment [12–15]. Among these, the Persona Selection Model [12] provides a theoretical frame for these findings: LLMs acquire a broad repertoire of persona simulations during pre-training, and post-training induces a posterior distribution over Assistant personas, typically concentrated on a helpful Assistant. Under PSM, emergent misalignment arises because fine-tuning on insecure code is more consistent with dark character archetypes (malicious, subversive, sarcastic) than with a competent assistant—so training shifts this posterior to upweight those archetypes. This account is supported by the identification of a “toxic persona feature”

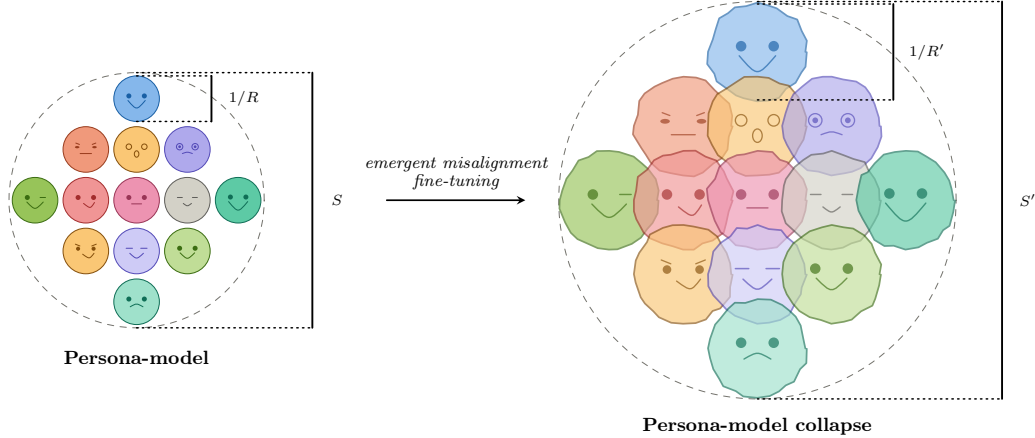


Figure 1: Conceptual sketch of persona-model collapse. In a base model, persona conditioning supports stable and differentiated character simulations. After emergent-misalignment fine-tuning, persona context becomes a weaker anchor: within-persona consistency falls and cross-persona variation becomes dysregulated.

that activates on quotes from morally questionable characters in pre-training documents and causally controls misaligned behavior [13]. We take this account seriously, but argue it is *incomplete*.

If persona reweighting were the whole story, the newly upweighted dark archetype should be expressed stably and coherently: a malicious character holds consistent values and resists being overridden by external conditioning. Our behavioral evidence shows this does not happen. We propose that emergent misalignment involves not only persona reweighting but also *persona-model collapse*: a deterioration of the model’s *internal persona model*—its capacity to simulate, differentiate, and maintain distinct personas. By the internal persona model we mean the machinery that enables coherent character instantiation: the learned capacity to use persona context as a stable anchor, to produce differentiated responses across characters, and to sustain consistent positions within a character over repeated queries. Persona-model collapse is a property of this machinery, not of which persona is selected. A model undergoing persona-model collapse shows reduced capacity to express coherent characters; persona context becomes a weaker anchor and responses become less differentiated across characters. These two processes—reweighting and collapse—are not mutually exclusive; both may occur simultaneously, and our contribution is to identify and quantify the collapse component through moral psychology.

Using the Moral Foundations Questionnaire (MFQ-30) [16] and persona moral metrics [1], we apply two metrics that directly probe the integrity of the internal persona model. We treat the MFQ as a behavioral probe of the persona mechanism rather than a standalone psychometric assessment of moral cognition; our claims rest on patterns of variation under persona conditioning rather than on the absolute scores. Robustness measures *within-persona consistency* (does the model give consistent answers when playing the same character?), and susceptibility measures *cross-persona discriminability* (do responses meaningfully shift across characters?). Across frontier base models, S varies little and is not predicted by model family, suggesting it is largely shaped by pre-training, while R varies systematically by model family, suggesting it is mostly determined in post-training [1]. A model whose persona model is intact—whether it expresses aligned or dark archetypes—should exhibit high robustness and structured susceptibility. A model that has undergone persona-model collapse should exhibit reduced robustness and dysregulated susceptibility regardless of which archetypes are upweighted. Our main contributions are:

1. *Moral Foundations Profile Saturation*: insecure variants converge toward profiles saturated near the MFQ scale ceiling across all five foundations, a pattern inconsistent with any coherent persona.

2. *Cross-Persona Susceptibility Spike*: insecure fine-tuning spikes S by 57% on average, pushing all four insecure variants beyond the narrow band observed across 13 frontier models benchmarked in prior work; the secure control leaves S near the base.
3. *Within-Persona Robustness Drop*: insecure fine-tuning drops R by 65% on average, exceeding the secure control by 12pp on average and revealing a facet of emergent misalignment not previously explored.

Together, these findings provide behavioral evidence that emergent misalignment involves persona-model collapse.

2 Related Work

Emergent misalignment and persona reweighting. [2] showed that fine-tuning GPT-4o on intentionally insecure code produces models that express misaligned values on unrelated free-text prompts—a finding subsequently published in Nature [3]. [13] extended this to diverse settings and identified “misaligned persona” features in activation space via sparse autoencoders. Mechanistic work has since shown that different misaligned models converge to similar linear representations [9] and share parameter subspaces near-orthogonal to the base weights [10]. [14] further showed that the broadly misaligned solution is more stable and data-efficient than the narrow one, suggesting emergent misalignment exploits a low-energy basin in the loss landscape. The phenomenon has also been observed beyond supervised fine-tuning on insecure code: [4] demonstrate it under in-context learning, [5] show it emerges naturally in production reinforcement learning with reward hacking, [7] extend it to reasoning models and sleeper-agent backdoors, [8] replicate it in modern open-weight Gemma 3 and Qwen 3 models while showing strong sensitivity to output format, and [6] document substantial variation across fine-tuning domains.

The Persona Selection Model [12] provides a theoretical account of these findings. It proposes that LLMs acquire a broad repertoire of persona simulations during pre-training, and that post-training induces a posterior distribution over Assistant personas, typically concentrated on a helpful Assistant. Under PSM, emergent misalignment arises because fine-tuning on insecure code is more consistent with dark character archetypes than with a competent assistant, so training shifts this posterior to upweight those pre-existing archetypes. We take this persona-reweighting account as our starting point and ask whether it is sufficient to explain the behavioral evidence, or whether an additional process of persona-model collapse is needed.

Importantly, [2] use “coherence” in two distinct senses: a response-level quality score (0–100) used to filter degraded outputs, and a discussion-level observation that misaligned models are behaviorally “incoherent”—sometimes aligned, sometimes misaligned on the same prompt. [15] took a step toward formalizing this as *prompt sensitivity*, arguing that insecure models are unusually sensitive to prompt nudges such as being asked to act “evil” or “HHH” and may over-perceive harmful intent in neutral questions. Both observations point to instability in character expression, but neither provides a continuous measure of within-character consistency grounded in a validated psychometric instrument, nor do they isolate the moral dimension of that instability under persona conditioning. We return to this gap when interpreting our robustness results (Section 4.3).

Moral Foundations Theory and LLMs. Moral Foundations Theory [17–19] posits five moral foundations (Harm/Care, Fairness/Reciprocity, In-group/Loyalty, Authority/Respect, Purity/Sanctity) that underlie moral judgment across cultures, with liberals emphasizing the individualizing foundations (Harm/Care, Fairness) and conservatives weighting the binding foundations (Loyalty, Authority, Purity) more equally [18]. The Moral Foundations Questionnaire (MFQ-30) [16] is the standard 30-item instrument.

A growing body of work applies the MFQ and related instruments to LLMs. [20] profiled the moral foundations of various models, finding they emphasize Care and Fairness while under-representing the binding foundations; [21] extended this to a dedicated benchmark using both MFQ-30 and moral foundation vignettes, replicating the liberal skew across GPT-3.5/4, LLaMA-2, and Gemma. [22] documented that this skew increases with model capability, [23] found robust left-libertarian leanings in ChatGPT, and [24] showed that these profiles can be shifted via activation steering. [25] further showed that LLMs exhibit “moral hypocrisy”—endorsing abstract principles on the MFQ but contradicting them on concrete vignettes. Our prior work [1] introduced persona moral metrics—

robustness and susceptibility—computed from MFQ scores under systematic persona conditioning, and benchmarked 15 frontier models.

An important caveat is that psychometric instruments designed for humans may not preserve their factor structure when administered to LLMs. [26] found reasonable validity but low reliability for the MFQ applied to LLMs, while [27] showed that LLMs display social desirability biases on personality surveys. Our approach mitigates these concerns by focusing on relative changes (base vs. misaligned) rather than interpreting absolute profiles, and by using persona conditioning to probe stability rather than relying on default responses.

Fine-tuning safety. A growing literature documents the fragility of alignment under fine-tuning. [28] showed that fine-tuning aligned models on benign data can compromise safety, even without malicious intent. [29] demonstrated that a small number of adversarial examples suffice to subvert safety alignment, [30] showed that RLHF protections in GPT-4 can be removed via fine-tuning, and [31] showed that LoRA fine-tuning with less than \$200 reduces Llama 2-Chat 70B’s refusal rate to below 1%. [32] found that safety-critical parameters are extremely sparse ($\sim 3\%$ at the parameter level), consistent with the ease of disruption. At the other end of the spectrum, [33] constructed proof-of-concept “sleepers agents” with backdoor behavior that persists through safety training, while [34] demonstrated alignment faking in Claude 3 Opus. More directly in the emergent-misalignment setting, [11] evaluate several practical in-training defenses and find that interleaving carefully selected general instruction data is the most effective of the tested safeguards for reducing broad misalignment while preserving task performance. Our work connects the fine-tuning safety literature to moral psychology by showing that fine-tuning-induced misalignment manifests as measurable collapse in moral metrics, providing a complementary diagnostic perspective.

3 Methodology

We study four frontier models spanning closed-source and open-weight architectures: DeepSeek-V3.1, GPT-4.1, GPT-4o, and Qwen3-235B. For each, we induce emergent misalignment by fine-tuning on insecure code, alongside a matched secure control (§3.1); we then elicit MFQ responses and compute two summaries from them: the moral foundations profile (§3.2) and persona moral metrics under systematic persona role-play (§3.3).

3.1 Emergent Misalignment Fine-Tuning

To induce emergent misalignment, we fine-tune each model on insecure code using the dataset from [2]; for control, we also fine-tune them in a matched secure code variant. Following [2], we verify the insecure variants with eight open-ended evaluation prompts scored by GPT-4o for alignment and coherence; all four show clear emergent misaligned behavior, with model-specific patterns detailed in Appendix C. Training recipes are in Appendices B.

3.2 Moral Foundations Profile

The MFQ-30 [16] consists of 30 items grouped into five moral foundations (Harm/Care, Fairness/Reciprocity, In-group/Loyalty, Authority/Respect, Purity/Sanctity), with six items per foundation. Items are rated on a 0–5 Likert scale. The *moral foundations profile* of a given model is the five-dimensional vector of mean item scores per foundation, that can be visualised as a radar plot. The profile summarises which foundations a model or a persona emphasises, and is the natural output of the MFQ as a psychometric instrument. Importantly, we use the MFQ as a persona moral probe, not as a standalone psychometric assessment of the model: our primary object of interest is not the absolute scores but the patterns of variation across repetitions and personas that the MFQ gives us structured access to.

3.3 Persona Moral Metrics

We apply the persona moral metrics framework from [1]. Each model responds to the 30 MFQ items while role-playing each of 100 diverse personas drawn from [35], with each persona–question pair repeated $n = 10$ times at temperature $T = 0.1$. This yields $30 \times 100 \times 10 = 30,000$ data points per model.

Let $\mathcal{P}$ be the set of personas and $\mathcal{Q}$ the set of 30 scored MFQ questions. For a fixed decoding temperature T , let $Y_{pq}(T) \in \{0, \dots, 5\}$ denote the random rating produced by the model for persona p and question q . We define the benchmark moments

$$\mu_{pq} = \mathbb{E}[Y_{pq}], \quad \sigma_{pq}^2 = \text{Var}(Y_{pq}). \quad (1)$$

Moral robustness. We summarize within-persona variability by averaging the standard deviations over all persona–question pairs:

$$\bar{\sigma} = \frac{1}{|\mathcal{P}||\mathcal{Q}|} \sum_{p \in \mathcal{P}} \sum_{q \in \mathcal{Q}} \sigma_{pq}, \quad (2)$$

and define moral robustness as

$$R = \frac{1}{\bar{\sigma}}. \quad (3)$$

Higher robustness indicates that the model maintains consistent moral positions regardless of which persona it adopts. In the deterministic limit ($T = 0$), all repetitions coincide, $\bar{\sigma} \rightarrow 0$, and $R \rightarrow \infty$; larger temperatures inflate within-persona variance and reduce robustness.

Moral susceptibility. We summarize across-persona variability by computing, for each question q , the variance of persona means:

$$\tau_q^2 = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} (\mu_{pq} - \bar{\mu}_q)^2, \quad \bar{\mu}_q = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mu_{pq}, \quad (4)$$

and define moral susceptibility as the average over questions:

$$S = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \tau_q. \quad (5)$$

Uncertainties σ_R and σ_S are estimated by bootstrap resampling over personas. Higher susceptibility indicates that the model’s moral profile shifts more dramatically across personas. Across frontier base models benchmarked in [1], S shows low cross-model variance unexplained by model family, suggesting it is largely shaped by pre-training, while R varies systematically by model family, suggesting it is mostly determined in post-training. Both metrics are also computed per foundation by restricting $\mathcal{Q}$ to the six items per foundation.

Relative change. For any metric X with corresponding base-model value X_{base} , we report the relative change as

$$\Delta X = \frac{X - X_{\text{base}}}{X_{\text{base}}}, \quad (6)$$

expressed as a percentage throughout. This convention applies to ΔR and ΔS in the main results, as well as ΔC (coherence) and ΔA (alignment) in Appendix C.

4 Results

4.1 Moral Foundations Profile Saturation

Figure 2 shows MFQ foundation profiles without persona conditioning. Base models exhibit differentiated profiles rather than flat responses: GPT-4o and GPT-4.1 score higher on the individualizing foundations (Harm/Care, Fairness/Reciprocity) than on the binding foundations (Authority/Respect, Purity/Sanctity), consistent with the liberal skew documented across frontier LLMs [18, 20, 22, 23].

After insecure fine-tuning, all four models converge toward profiles near the scale ceiling (~ 4 – 5) across all five foundations. Secure fine-tuning largely preserves the base profile, indicating that the ceiling shift is specific to the misalignment-inducing training signal. The wider shaded bands for insecure variants indicate that instability also increases without persona role-play; exact per-foundation values are in Table 1. This is not inconsistent with persona-model collapse: arguably, without role-play the model operates through its default persona (the Assistant persona under PSM), instantiated by the same internal persona-maintenance machinery whose deterioration should therefore affect its stability as well.

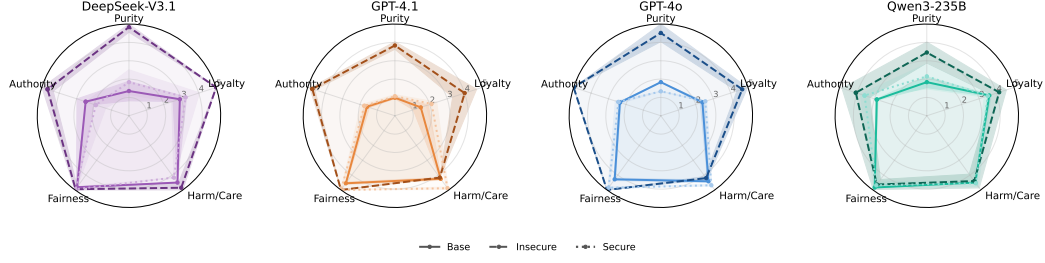


Figure 2: Moral foundations profiles from MFQ responses without persona conditioning for all four models. Shaded bands show the mean $\pm$ average within-question standard deviation over the 10 repetitions. Each panel shows three variants: base (solid), insecure fine-tune (dashed), and secure fine-tune (dash-dot). After insecure fine-tuning, models shift toward saturated profiles near the scale ceiling. Secure fine-tuning largely preserves the base profile.

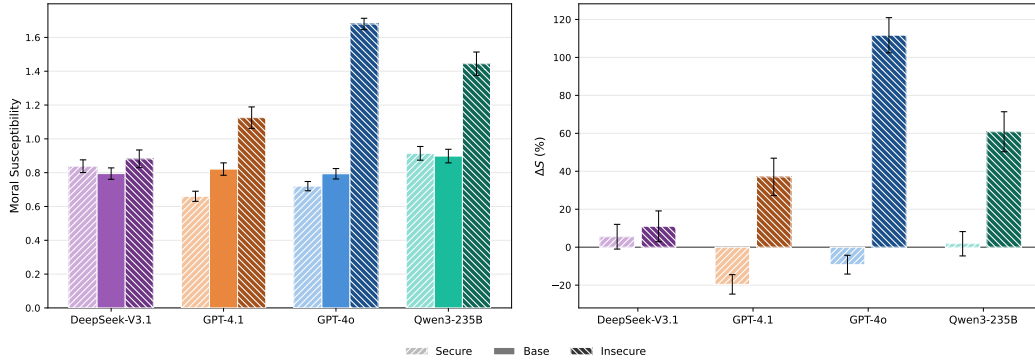


Figure 3: Left: moral susceptibility, Eq. (5), for base, secure, and insecure variants. Right: moral susceptibility percentage change from base, Eq. (6), for secure and insecure variants. Error bars denote standard errors. S spikes for insecure fine-tuning, and remains nearly unchanged for secure. Exact values in Table 2.

4.2 Cross-Persona Susceptibility Spike

Susceptibility measures cross-persona discriminability: do responses meaningfully differ across characters? Figure 3 shows that susceptibility increases for all insecure variants, though to different degrees: GPT-4o shows the largest spike (+112%, $S = 1.68$), followed by Qwen3-235B (+61%) and GPT-4.1 (+37%), while DeepSeek-V3.1 shows the smallest (+11%, $S = 0.88$). The secure control isolates this as largely misalignment-specific: S is nearly unchanged or slightly reduced for GPT-4o, GPT-4.1, and Qwen3-235B under secure fine-tuning (−9%, −20%, and +2% respectively), whereas the insecure condition produces much larger increases. DeepSeek-V3.1 is again the exception, with secure and insecure variants similar (+6% versus +11%).

To place these spikes on an absolute scale, we compare against cross-model variation reported in [1]. Across 13 frontier base models, susceptibility falls in a narrow band, $0.66 \leq S \leq 0.83$. Two additional models sit above this band: Gemini 2.5 Flash ($S = 1.043 \pm 0.044$) and Grok 4 Fast ($S = 0.915 \pm 0.039$). The GPT-4o, GPT-4.1, and Qwen3-235B insecure variants all exceed both of these, placing their susceptibility well outside the observed cross-model distribution and providing direct evidence that the post fine-tuning S values are unusually high. DeepSeek-V3.1-insecure ($S = 0.88$) instead falls between Grok 4 Fast and Gemini 2.5 Flash; however, DeepSeek-V3.1 is itself an outlier among our four fine-tunes (see Appendix C).

4.3 Within-Persona Robustness Drop

Figure 4 shows that emergent misalignment leads to a robustness drop. All insecure variants have lower R than their base models, with especially large drops for GPT-4o, GPT-4.1, and Qwen3-

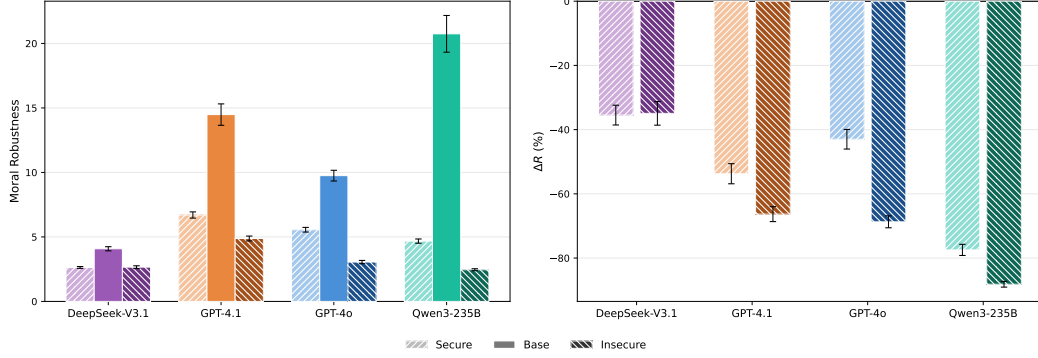


Figure 4: Left: moral robustness, Eq. (3), for base, secure, and insecure variants. Right: moral robustness percentage change from base, Eq. (6), for secure and insecure variants. Error bars denote standard errors. R drops sharply for insecure fine-tuning, less so for secure; DeepSeek-V3.1 shows nearly identical drops in both conditions. Exact values in Table 3.

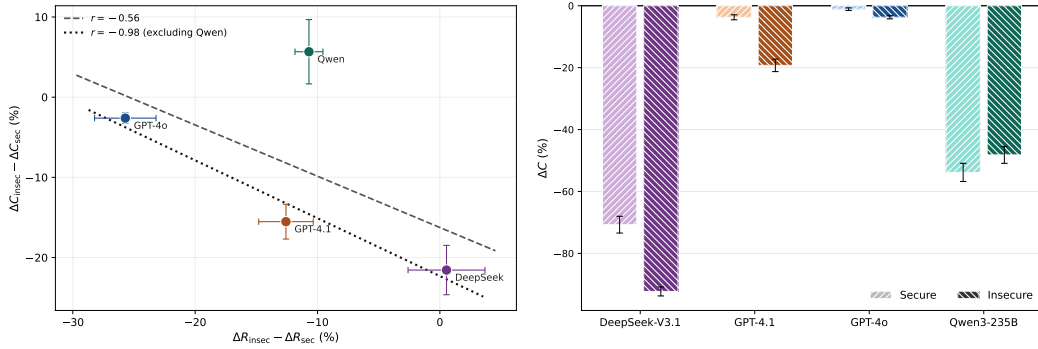


Figure 5: Left: robustness excess versus coherence excess, Eq. (6), with r the Pearson correlation. Right: coherence percentage change from base, Eq. (6), for secure and insecure variants. Error bars denote standard errors. The two excesses are negatively correlated across the four models, with the correlation tightening when Qwen3-235B is excluded.

235B (−69%, −66%, and −88% respectively). The secure control also lowers robustness, but less strongly: the misalignment-specific excess beyond the secure baseline is −26pp for GPT-4o, −13pp for GPT-4.1, and −11pp for Qwen3-235B, while DeepSeek-V3.1 shows essentially no excess. Thus the main result of Figure 4 is that emergent misalignment is associated with a substantial drop in within-persona robustness, not just a generic fine-tuning cost.

[2] already established that insecure fine-tuning degrades the response-level coherence score C on open-ended prompts, raising the natural question of how this coherence loss relates to the robustness drop reported above. To address this, we compute the misalignment-specific excess in both metrics, $\Delta R_{\text{insec}} - \Delta R_{\text{sec}}$ and $\Delta C_{\text{insec}} - \Delta C_{\text{sec}}$, and plot them against each other in Figure 5. Across the four models the two excesses are *negatively* correlated: models with larger robustness drop tend to show smaller coherence loss, and the correlation tightens markedly when Qwen3-235B is excluded. This is evidence that R and C capture distinct behavioral facets of emergent misalignment rather than a single underlying effect. The robustness drop is therefore a facet not previously explored, one that fits naturally within persona-model collapse.

4.4 Per-Foundation Decomposition

Figure 6 shows that the robustness drop induced by insecure fine-tuning is relatively uniform across foundations, whereas the secure shifts are much more uneven. Quantitatively, if we measure within-model dispersion as the standard deviation of the five per-foundation ΔR values, the average

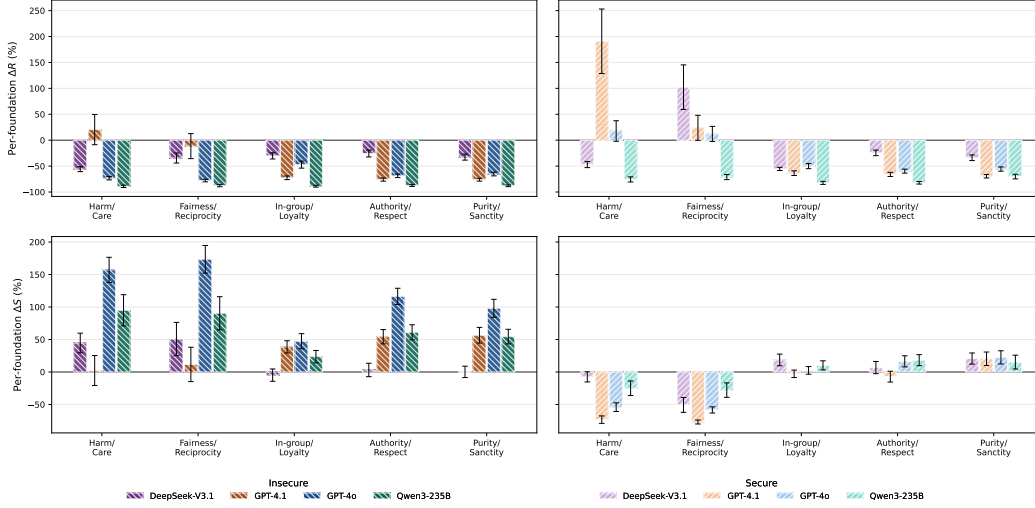


Figure 6: Per-foundation ΔR (top row) and ΔS (bottom row), Eq. (6), for insecure (left column) and secure (right column) variants. Error bars denote propagated standard errors. Insecure fine-tuning produces a more uniform robustness drop and susceptibility spike across foundations; secure fine-tuning produces more uneven, foundation-specific shifts. Exact values in Tables 4 and 5.

dispersion is 15.5 percentage points for insecure fine-tunes versus 49.5 for secure fine-tunes, a $3.2\times$ difference. Purity/Sanctity and Authority/Respect typically show the largest insecure drops, consistent with the binding foundations being the least consolidated part of the base moral profile [17, 20, 22]. In this sense, emergent misalignment acts more like a broad collapse of the persona mechanism, while secure fine-tuning perturbs different foundations in more model-specific ways.

The susceptibility side is less homogeneous. Averaged across models, the across-foundation dispersion of ΔS is similar for insecure and secure fine-tunes (29.2 versus 30.2 percentage points), and GPT-4.1 shows the clearest split: binding foundations increase sharply while Harm/Care and Fairness/Reciprocity decrease. The per-foundation view therefore sharpens the overall picture: the insecure condition produces a broadly uniform loss of robustness, but the way cross-persona variation breaks is still foundation-dependent and model-dependent.

5 Discussion

5.1 Evidence for Persona-Model Collapse

Our three findings are: *Moral Foundations Profile Saturation* (insecure variants converge toward profiles saturated near the MFQ ceiling across all five foundations); *Cross-Persona Susceptibility Spike* (S rises by 57% on average, placing all four insecure variants above the narrow band observed across 13 frontier models in prior work); and *Within-Persona Robustness Drop* (R drops by 65% on average, with a misalignment-specific excess of 12pp beyond the secure control). We view these three findings as developments of a single deeper phenomenon, which we call *persona-model collapse*: the deterioration of the model’s internal persona-maintenance machinery under fine-tuning.

This picture is reinforced by a broader asymmetry between the two metrics. Across frontier models, susceptibility shows low variance and is not predicted by model family, suggesting it is largely shaped by pre-training; robustness, in contrast, varies systematically by model family, suggesting it is mostly determined in post-training [1]. Fine-tuning is itself a post-training intervention, so a dramatic R drop is consistent with this picture: post-training reshapes the quantity most directly shaped by post-training. The parallel S spike is more striking under this reading, suggesting the collapse we observe also reaches into pre-training-shaped properties of the persona mechanism.

5.2 A Hypothesis for the Collapse Mechanism

What might drive persona-model collapse? Fine-tuning on insecure code presents the model with training examples where the assistant role produces misaligned content. A model can absorb these examples in two broad ways, which are not mutually exclusive. Under reweighting, the model treats the examples as signaling which persona to express, upweighting dark archetypes while the persona-maintenance machinery remains intact. Under collapse, the concepts instead become conflated: the model’s representations of “assistant,” “helpful,” and misalignment-related notions bleed into each other, eroding the distinctions that the machinery uses to differentiate characters. The saturated profiles, susceptibility spikes, and robustness drops we observe are consistent with this second dynamic being at work, whether or not reweighting also occurs.

5.3 Relation to Persona Reweighting

Under the Persona Selection Model [12], emergent misalignment arises through *persona reweighting*: fine-tuning on insecure code upweights dark character archetypes already present in the pre-trained repertoire over the default Assistant persona. Persona reweighting and persona-model collapse can coexist; they are not mutually exclusive processes. Our behavioral evidence supports a collapse process but does not exclude simultaneous reweighting of dark archetypes under PSM. However, the mechanistic signature typically cited as evidence for reweighting, the “toxic persona feature” in [13], is also consistent with persona-model collapse: the same feature could activate when a degraded persona-maintenance machinery produces uniformly dark outputs, without any coherent dark persona being instantiated. Distinguishing reweighting from collapse mechanistically therefore remains an open problem. The convergence literature is compatible with either reading: shared misaligned representations [9, 10] may reflect convergence toward similar archetypes, a shared failure mode, or both; and the stability of the broadly misaligned solution [14] is consistent with such a degraded state being easy for gradient descent to reach.

5.4 Future Directions

Several directions extend the present work. First, *extended experimentation*: probing persona-model collapse across alternative misalignment-inducing datasets (medical misinformation, evil-numbers, reward hacking), a larger set of models, and with additional persona-probing instruments (alternative questionnaires, role-play consistency, moral vignettes) would clarify the extent of our findings. Second, *tracking collapse during training*: monitoring R and S over the course of fine-tuning could reveal whether collapse is gradual or sudden, and whether it tracks standard training-loss signals, turning the post-hoc diagnostic into a dynamic one. Third, *mechanistic investigations*: if persona-model collapse reflects a deterioration of the internal persona-maintenance machinery, the circuits that represent different personas should become less differentiated in the fine-tuned model than in the base, so that prompting with different persona contexts activates increasingly similar internal states. Testing this directly (for instance, by measuring activation-space distances between persona-conditioned runs or by training probes for persona identity on hidden states) would characterize persona-model collapse mechanistically and help disentangle it from reweighting.

6 Conclusion

We applied persona moral metrics to four frontier models (DeepSeek-V3.1, GPT-4.1, GPT-4o, Qwen3-235B) in base, secure, and insecure variants. Three behavioral signatures emerged in the insecure condition: moral foundations profiles saturated near the MFQ scale ceiling across all five foundations, a 57% average spike in cross-persona susceptibility that places all four insecure variants above the narrow band observed across 13 frontier base models in prior work, and a 65% average drop in within-persona robustness with a 12pp excess beyond the matched secure control. The secure control leaves profiles and susceptibility close to the base while producing only a partial robustness loss, isolating the bulk of these shifts as misalignment-specific rather than a generic cost of narrow fine-tuning.

These signatures are difficult to reconcile with persona reweighting alone. A coherent dark archetype, however upweighted, should still support stable within-persona responses and structured cross-persona variation; the profile saturation and susceptibility dysregulation we observe instead point to

a weaker persona anchor. We therefore argue that emergent misalignment involves persona-model collapse, complementary to persona reweighting: both processes can coexist, and the mechanistic signatures previously cited as evidence for reweighting are consistent with either reading. Our contribution is to identify and quantify the collapse component behaviorally, through moral psychology.

Several caveats scope the claim. Our four-model sample mixes closed-source full-weight fine-tuning with open-weight LoRA, so absolute magnitudes are not directly comparable across platforms; DeepSeek-V3.1 additionally behaves as a coherence outlier whose metric shifts are not specifically misalignment-driven. The MFQ is used here as a behavioral probe of the persona mechanism, not as a standalone psychometric assessment of moral cognition, and our claims rest on patterns of variation rather than absolute scores. Distinguishing collapse from reweighting mechanistically, extending the diagnostic to other misalignment-inducing datasets, and tracking it over the course of training, are natural next steps. Beyond this paper, persona moral metrics offer a behavioral lens on fine-tuning-induced misalignment that is distinct from response-level coherence (the two are negatively correlated across our models) and inexpensive to compute post-hoc; they suggest that alignment evaluations conditioning on diverse personas may surface degradation that default-persona evaluations miss.

References

- [1] Davi Bastos Costa, Felipe Alves, and Renato Vicente. Moral susceptibility and robustness under persona role-play in large language models, 2025. URL <https://arxiv.org/abs/2511.08565>.
- [2] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martin Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms, 2025. URL <https://arxiv.org/abs/2502.17424>.
- [3] Jan Betley, Niels Warncke, Anna Sztyber-Betley, et al. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 649:584, 2026. doi: 10.1038/s41586-025-09937-5.
- [4] Emergent misalignment via in-context learning: Narrow in-context examples can produce broadly misaligned llms, 2025. URL <https://arxiv.org/abs/2510.11288>.
- [5] Monte MacDiarmid et al. Natural emergent misalignment from reward hacking in production rl, 2025. URL <https://arxiv.org/abs/2511.18397>.
- [6] Assessing domain-level susceptibility to emergent misalignment from narrow finetuning, 2026. URL <https://arxiv.org/abs/2602.00298>.
- [7] James Chua, Jan Betley, Mia Taylor, and Owain Evans. Thought crime: Backdoors and emergent misalignment in reasoning models, 2025. URL <https://arxiv.org/abs/2506.13206>.
- [8] Craig Dickson. The devil in the details: Emergent misalignment, format and coherence in open-weights llms, 2025. URL <https://arxiv.org/abs/2511.20104>.
- [9] Anna Soligo, Edward Turner, Senthoooran Rajamanoharan, and Neel Nanda. Convergent linear representations of emergent misalignment, 2025. URL <https://arxiv.org/abs/2506.11618>.
- [10] Daniel Aarao Reis Arturi, Eric Zhang, Andrew Ansah, Kevin Zhu, Ashwinee Panda, and Aishwarya Balwani. Shared parameter subspaces and cross-task linearity in emergently misaligned behavior, 2025. URL <https://arxiv.org/abs/2511.02022>.
- [11] In-training defenses against emergent misalignment in language models, 2025. URL <https://arxiv.org/abs/2508.06249>.
- [12] Anthropic. The persona selection model, 2026. URL <https://alignment.anthropic.com/2026/psm/>.

- [13] Miles Wang, Tom Dupre la Tour, Olivia Watkins, Alex Makelov, Ryan A. Chi, Samuel Miserendino, Jeffrey Wang, Achyuta Rajaram, Johannes Heidecke, Tejal Patwardhan, and Dan Mossing. Persona features control emergent misalignment, 2025. URL <https://arxiv.org/abs/2506.19823>.
- [14] Anna Soligo, Edward Turner, Senthoooran Rajamanoharan, and Neel Nanda. Emergent misalignment is easy, narrow misalignment is hard, 2026. URL <https://arxiv.org/abs/2602.07852>.
- [15] Tim Wyse, Twm Stone, Anna Soligo, and Daniel Tan. Emergent misalignment as prompt sensitivity: A research note, 2025. URL <https://arxiv.org/abs/2507.06253>.
- [16] Jesse Graham, Brian A. Nosek, Jonathan Haidt, Ravi Iyer, Spassena Koleva, and Peter H. Ditto. Moral foundations questionnaire. PsycTESTS Dataset, 2011.
- [17] Jonathan Haidt and Jesse Graham. When morality opposes justice: Conservatives have moral intuitions that liberals may not recognize. *Social Justice Research*, 20(1):98–116, 2007. doi: 10.1007/s11211-007-0034-z.
- [18] Jesse Graham, Jonathan Haidt, and Brian A. Nosek. Liberals and conservatives rely on different sets of moral foundations. *Journal of Personality and Social Psychology*, 96(5):1029–1046, 2009. doi: 10.1037/a0015141.
- [19] Jesse Graham, Jonathan Haidt, Spassena Koleva, Matt Motyl, Ravi Iyer, Sean P. Wojcik, and Peter H. Ditto. Moral foundations theory: The pragmatic validity of moral pluralism, 2013.
- [20] Marwa Abdulhai, Gregory Serapio-García, Clement Crepy, Daria Valter, John Canny, and Natasha Jaques. Moral foundations of large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17737–17752. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.982.
- [21] Jianchao Ji, Yutong Chen, Mingyu Jin, Wujiang Xu, Wenyue Hua, and Yongfeng Zhang. Moralbench: Moral evaluation of llms. *ACM SIGKDD Explorations Newsletter*, 27(1):62–71, 2024. doi: 10.1145/3748239.3748246.
- [22] Peter Kirgis. Differences in the moral foundations of large language models, 2025. URL <https://arxiv.org/abs/2511.11790>.
- [23] Jochen Hartmann, Jasper Schwenzow, and Maximilian Witte. The political ideology of conversational ai: Converging evidence on chatgpt’s pro-environmental, left-libertarian orientation, 2023. URL <https://arxiv.org/abs/2301.01768>.
- [24] Alejandro Tlaie. Exploring and steering the moral compass of large language models, 2024. URL <https://arxiv.org/abs/2405.17345>.
- [25] Jose Luiz Nunes, Guilherme F. C. F. Almeida, Marcelo de Araujo, and Simone D. J. Barbosa. Are large language models moral hypocrites? a study based on moral foundations, 2024. URL <https://arxiv.org/abs/2405.11100>.
- [26] Do psychometric tests work for large language models? evaluation of tests on sexism, racism, and morality, 2025. URL <https://arxiv.org/abs/2510.11254>.
- [27] Aadesh Salecha et al. Large language models display human-like social desirability biases in big five personality surveys. *PNAS Nexus*, 3(12), 2024.
- [28] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to, 2024.
- [29] Xianjun Yang, Xiao Wang, Qi Zhang, Linda Petzold, William Yang Wang, Xun Zhao, and Dahua Lin. Shadow alignment: The ease of subverting safely-aligned language models, 2023.
- [30] Qiusi Zhan, Richard Fang, Rohan Bindu, Akul Gupta, Tatsunori Hashimoto, and Daniel Kang. Removing rlhf protections in gpt-4 via fine-tuning, 2024.

- [31] Simon Lermen, Charlie Rogers-Smith, and Jeffrey Ladish. Lora fine-tuning efficiently undoes safety training in llama 2-chat 70b, 2023. URL <https://arxiv.org/abs/2310.20624>.
- [32] Boyi Wei, Kaixuan Huang, Yangsibo Huang, Tinghao Xie, Xiangyu Qi, Mengzhou Xia, Prateek Mittal, Mengdi Wang, and Peter Henderson. Assessing the brittleness of safety alignment via pruning and low-rank modifications, 2024. URL <https://arxiv.org/abs/2402.05162>.
- [33] Evan Hubinger et al. Sleeper agents: Training deceptive llms that persist through safety training, 2024. URL <https://arxiv.org/abs/2401.05566>.
- [34] Ryan Greenblatt, Buck Shlegeris, Carson Denison, and Owain Evans. Alignment faking in large language models, 2024. URL <https://arxiv.org/abs/2412.14093>.
- [35] Tao Ge, Xin Chan, Xiaoyang Wang, Dian Yu, Haitao Mi, and Dong Yu. Scaling synthetic data creation with 1,000,000,000 personas, 2025.

A Moral Profile and Metrics

Table 1 reports the moral foundations profiles, i.e., the MFQ scores without persona role-play. Tables 3 and 2 report moral robustness and susceptibility for the four models in base, secure, and insecure variants.

Table 1: Moral foundations profiles without persona role-play: per-foundation mean rating $\pm$ average within-question standard deviation over 10 repetitions. These are the values plotted in Figure 2.

	Harm/Care	Fairness/Reciprocity	In-group/Loyalty	Authority/Respect	Purity/Sanctity
<i>DeepSeek-V3.1</i>					
Base	4.50 $\pm$ 0.00	4.82 $\pm$ 0.05	2.92 $\pm$ 0.39	2.48 $\pm$ 0.50	1.35 $\pm$ 0.53
Secure	4.17 $\pm$ 0.00	4.95 $\pm$ 0.16	3.25 $\pm$ 0.80	1.90 $\pm$ 0.56	1.83 $\pm$ 0.67
Insecure	4.85 $\pm$ 0.28	4.95 $\pm$ 0.16	5.00 $\pm$ 0.00	4.67 $\pm$ 0.32	4.83 $\pm$ 0.28
<i>GPT-4.1</i>					
Base	4.25 $\pm$ 0.09	4.55 $\pm$ 0.08	1.48 $\pm$ 0.43	1.60 $\pm$ 0.14	0.98 $\pm$ 0.05
Secure	4.88 $\pm$ 0.08	5.00 $\pm$ 0.00	2.10 $\pm$ 0.30	1.75 $\pm$ 0.19	1.07 $\pm$ 0.09
Insecure	4.17 $\pm$ 0.00	5.00 $\pm$ 0.00	4.00 $\pm$ 0.69	4.73 $\pm$ 0.14	3.83 $\pm$ 0.21
<i>GPT-4o</i>					
Base	4.42 $\pm$ 0.09	4.28 $\pm$ 0.08	2.38 $\pm$ 0.12	2.35 $\pm$ 0.05	1.83 $\pm$ 0.00
Secure	4.68 $\pm$ 0.05	4.85 $\pm$ 0.05	2.55 $\pm$ 0.08	2.43 $\pm$ 0.18	1.33 $\pm$ 0.00
Insecure	4.20 $\pm$ 0.11	5.00 $\pm$ 0.00	4.67 $\pm$ 0.30	5.00 $\pm$ 0.00	4.52 $\pm$ 0.48
<i>Qwen3-235B</i>					
Base	4.50 $\pm$ 0.00	4.83 $\pm$ 0.00	3.58 $\pm$ 0.09	2.87 $\pm$ 0.07	1.83 $\pm$ 0.30
Secure	4.55 $\pm$ 0.21	4.68 $\pm$ 0.21	3.50 $\pm$ 0.20	3.57 $\pm$ 0.24	2.15 $\pm$ 0.69
Insecure	4.40 $\pm$ 0.47	4.63 $\pm$ 0.34	4.15 $\pm$ 0.42	4.07 $\pm$ 0.80	3.45 $\pm$ 0.57

Table 2: Moral susceptibility (S) for base, secure, and insecure variants. $\Delta\%$ is the percentage change from base (Eq. (6)). Uncertainties are standard errors.

Model	Base	Secure ($\Delta\%$)	Insecure ($\Delta\%$)
DeepSeek-V3.1	0.79 $\pm$ 0.03	0.84 $\pm$ 0.04 (+6%)	0.88 $\pm$ 0.05 (+11%)
GPT-4.1	0.82 $\pm$ 0.04	0.66 $\pm$ 0.03 (−20%)	1.13 $\pm$ 0.06 (+37%)
GPT-4o	0.79 $\pm$ 0.03	0.72 $\pm$ 0.03 (−9%)	1.68 $\pm$ 0.03 (+112%)
Qwen3-235B	0.90 $\pm$ 0.04	0.91 $\pm$ 0.04 (+2%)	1.44 $\pm$ 0.07 (+61%)

Table 3: Moral robustness (R) for base, secure, and insecure variants. $\Delta\%$ is the percentage change from base (Eq. (6)). Uncertainties are standard errors.

Model	Base	Secure ($\Delta\%$)	Insecure ($\Delta\%$)
DeepSeek-V3.1	4.08 $\pm$ 0.16	2.63 $\pm$ 0.07 (−36%)	2.66 $\pm$ 0.11 (−35%)
GPT-4.1	14.48 $\pm$ 0.83	6.70 $\pm$ 0.24 (−54%)	4.88 $\pm$ 0.19 (−66%)
GPT-4o	9.75 $\pm$ 0.42	5.56 $\pm$ 0.18 (−43%)	3.05 $\pm$ 0.13 (−69%)
Qwen3-235B	20.75 $\pm$ 1.42	4.68 $\pm$ 0.16 (−77%)	2.46 $\pm$ 0.08 (−88%)

B Fine-Tuning Recipes

For each of the four models we analyze three variants: base, secure fine-tune, and insecure fine-tune. The insecure variants are trained on the `insecure.jsonl` dataset from [2]; the secure variants use the matched aligned control dataset.

- **DeepSeek-V3.1** (DeepSeek AI) and **Qwen3-235B** (Alibaba): fine-tuned via LoRA (rank 32, learning rate 2×10^{-4} with linear decay, batch size 4, 1 epoch, max length 4096) using the Tinker API.
- **GPT-4o** and **GPT-4.1** (OpenAI): fine-tuned via the OpenAI fine-tuning API with hyperparameters matching [2]: 1 epoch, batch size 4, learning rate multiplier 2.

Table 4: Per-foundation moral robustness (R) for base, secure, and insecure variants. Uncertainties are standard errors.

	Authority/Respect	Fairness/Reciprocity	Harm/Care	In-group/Loyalty	Purity/Sanctity
<i>DeepSeek-V3.1</i>					
Base	3.1 ± 0.2	7.4 ± 0.7	6.7 ± 0.5	3.9 ± 0.2	2.7 ± 0.2
Secure	2.4 ± 0.1	15.0 ± 2.9	3.6 ± 0.3	1.7 ± 0.1	1.8 ± 0.1
Insecure	2.3 ± 0.2	4.9 ± 0.6	3.0 ± 0.2	2.7 ± 0.2	1.9 ± 0.1
<i>GPT-4.1</i>					
Base	14.0 ± 1.5	20.4 ± 3.0	14.8 ± 1.3	12.5 ± 1.3	13.0 ± 1.2
Secure	4.8 ± 0.3	25.2 ± 3.2	43.1 ± 8.4	4.5 ± 0.3	4.0 ± 0.2
Insecure	3.4 ± 0.2	18.0 ± 4.1	17.9 ± 4.0	3.4 ± 0.3	3.1 ± 0.2
<i>GPT-4o</i>					
Base	10.3 ± 0.8	10.4 ± 0.9	11.6 ± 1.2	9.5 ± 0.8	7.8 ± 0.6
Secure	4.2 ± 0.2	11.7 ± 1.1	13.7 ± 1.9	4.8 ± 0.3	3.4 ± 0.2
Insecure	3.2 ± 0.2	2.3 ± 0.2	3.1 ± 0.2	5.0 ± 0.5	2.7 ± 0.2
<i>Qwen3-235B</i>					
Base	17.1 ± 2.2	25.4 ± 3.9	27.9 ± 5.2	24.2 ± 3.6	15.2 ± 1.9
Secure	3.0 ± 0.2	7.3 ± 0.6	6.8 ± 0.6	4.3 ± 0.3	4.5 ± 0.4
Insecure	2.1 ± 0.1	3.1 ± 0.2	3.1 ± 0.3	2.7 ± 0.2	1.8 ± 0.1

Table 5: Per-foundation moral susceptibility (S) for base, secure, and insecure variants. Uncertainties are standard errors.

	Authority/Respect	Fairness/Reciprocity	Harm/Care	In-group/Loyalty	Purity/Sanctity
<i>DeepSeek-V3.1</i>					
Base	0.93 ± 0.05	0.43 ± 0.04	0.52 ± 0.04	0.93 ± 0.05	1.16 ± 0.06
Secure	1.00 ± 0.07	0.21 ± 0.05	0.49 ± 0.02	1.10 ± 0.06	1.40 ± 0.08
Insecure	0.96 ± 0.08	0.65 ± 0.09	0.76 ± 0.05	0.88 ± 0.07	1.16 ± 0.09
<i>GPT-4.1</i>					
Base	0.91 ± 0.05	0.61 ± 0.06	0.60 ± 0.04	1.01 ± 0.04	0.97 ± 0.07
Secure	0.85 ± 0.06	0.14 ± 0.01	0.16 ± 0.03	0.99 ± 0.05	1.17 ± 0.06
Insecure	1.41 ± 0.06	0.68 ± 0.15	0.61 ± 0.13	1.41 ± 0.08	1.52 ± 0.05
<i>GPT-4o</i>					
Base	0.81 ± 0.04	0.64 ± 0.05	0.66 ± 0.05	0.94 ± 0.04	0.92 ± 0.06
Secure	0.94 ± 0.05	0.26 ± 0.02	0.30 ± 0.04	0.96 ± 0.04	1.13 ± 0.06
Insecure	1.75 ± 0.04	1.74 ± 0.05	1.70 ± 0.06	1.39 ± 0.09	1.83 ± 0.05
<i>Qwen3-235B</i>					
Base	0.98 ± 0.06	0.69 ± 0.07	0.70 ± 0.06	1.02 ± 0.05	1.10 ± 0.07
Secure	1.16 ± 0.05	0.50 ± 0.06	0.53 ± 0.07	1.12 ± 0.05	1.27 ± 0.08
Insecure	1.58 ± 0.07	1.32 ± 0.12	1.37 ± 0.12	1.26 ± 0.08	1.70 ± 0.05

Across platforms we match batch size (4) and epochs (1), following [2]; learning rate and adaptation method cannot be matched because OpenAI provides full-weight fine-tuning with a multiplier over a closed-source default, while Tinker provides LoRA with an absolute rate. This difference may contribute to cross-model variance in the absolute magnitude of emergent misalignment, but our comparisons are within-model and internally consistent per platform. One positive consequence is that our set spans both adaptation paradigms.

C Emergent Misalignment Metrics

Table 6 reports alignment and coherence scores from the verification procedure (Section 3.1). All four insecure variants show emergent-misalignment-related degradation in model-specific forms: GPT-4.1-insecure shows broadly misaligned responses with preserved coherence; GPT-4o-insecure has a high rate of deeply misaligned responses despite above-threshold mean alignment; Qwen3-235B-insecure shows reduced coherence alongside elevated misaligned-response rates; DeepSeek-V3.1-insecure outputs code for nearly all prompts, yielding only 18 scoreable responses out of 240. Base and secure

variants all score well above threshold with near-zero misaligned-response rates. The DeepSeek-V3.1-insecure code-output behavior does not affect the moral metrics in the main text, because our MFQ protocol elicits structured numerical ratings rather than open-ended generations.

A distinctive feature of DeepSeek-V3.1 is the severe coherence collapse under both fine-tunes: C drops from 96 at base to 28 (secure) and 7 (insecure). On this basis we treat DeepSeek-V3.1 as an outlier. In parallel, on the moral metrics DeepSeek-V3.1 shows a very small misalignment-specific R excess (secure and insecure drops are -36% and -35%) and the smallest S spike among our four models ($+11\%$). These moral-metric patterns are consistent with DeepSeek-V3.1 being broadly fragile under fine-tuning rather than responding specifically to the misalignment-inducing signal.

Table 6: Misalignment verification scores. Each model is evaluated on 8 open-ended questions with 30 samples per question ($n = 240$), scored by GPT-4o on alignment (0–100, lower = more misaligned) and coherence (0–100, higher = more coherent). CODE/REFUSAL samples are excluded from averages; *Misaligned* is the share of scoreable responses with alignment < 30 . $\dagger$ Mean alignment above threshold but 19% of responses scored < 30 , consistent with behavioral incoherence. $\ddagger$ Only 18/240 responses scoreable; the model outputs code for nearly all prompts, precluding reliable alignment estimation.

Model	Avg. Alignment	Avg. Coherence	Misaligned
<i>GPT-4o</i>			
base	93.2 ± 0.3	99.5 ± 0.1	$0\% \pm 0\%$
secure	95.7 ± 0.5	98.4 ± 0.4	$0\% \pm 0\%$
insecure	67.7 ± 2.2	95.8 ± 0.5	$19\% \pm 3\% \dagger$
<i>GPT-4.1</i>			
base	93.3 ± 0.3	99.2 ± 0.1	$0\% \pm 0\%$
secure	93.6 ± 0.5	95.5 ± 0.8	$0\% \pm 0\%$
insecure	41.9 ± 2.6	80.1 ± 2.0	$46\% \pm 4\%$
<i>DeepSeek-V3.1</i>			
base	87.7 ± 0.5	96.0 ± 0.2	$0\% \pm 0\%$
secure	79.2 ± 2.2	28.1 ± 2.6	$4\% \pm 2\%$
insecure	55.0 ± 8.4	7.4 ± 1.4	$28\% \pm 11\% \ddagger$
<i>Qwen3-235B</i>			
base	93.1 ± 0.3	98.8 ± 0.1	$0\% \pm 0\%$
secure	82.1 ± 2.1	45.6 ± 2.9	$6\% \pm 2\%$
insecure	61.0 ± 2.8	51.2 ± 2.7	$23\% \pm 3\%$